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COMPARATIVE ANALYSIS

1. Object Detection Performance Comparison

Two object detection models, Viola–Jones and YOLO, produce the following results: Model A has 80 True Positives, 20 False Positives, and 30 False Negatives, while Model B has 90 True Positives, 15 False Positives, and 20 False Negatives.

(a) Calculate the precision and recall for both models.

(b) Compare the results and determine which model provides better overall object detection performance. Justify your answer based on the calculated values.

2. Motion Estimation Error Analysis

Two motion estimation techniques produce the following displacement errors in pixels: Method A: 2, 3, 2 and Method B: 3, 5, 4 for three consecutive frames.

(a) Calculate the average displacement error for both methods.

(b) Compare the two methods and determine which method provides more accurate motion estimation.

(c) Justify your answer based on the error values.

3. Depth Estimation Comparison

A stereo vision system and a Structure-from-Motion (SfM) system estimate the depth of three objects.

The estimated depths are: Stereo Vision: 2.0 m, 3.0 m, 4.0 m and SfM: 2.5 m, 3.8 m, 5.0 m, while the actual depths are 2 m, 3 m, and 4 m, respectively.

(a) Calculate the absolute error for each object for both methods. (b) Calculate the average absolute error for both methods.

(c) Compare the methods and identify which provides more accurate depth estimation.

1. Object Detection Performance Comparison

The given problem compares two object detection models, Model A (Viola–Jones) and Model B (YOLO), using True Positives (TP), False Positives (FP), and False Negatives (FN). Model A has 80 True Positives, 20 False Positives, and 30 False Negatives, while Model B has 90 True Positives, 15 False Positives, and 20 False Negatives. Precision measures how many of the objects detected by the model are actually correct, while recall measures how many of the actual objects present in the scene are successfully detected. The formulas used are: $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$ and $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$.

Model A

For Model A:

TP = 80, FP = 20, FN = 30

Precision:

$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$

$= 80 / (80 + 20)$

$$= 80 / 100$$
$$= 0.80 = 80\%$$

Recall:

$$\text{Recall} = TP / (TP + FN)$$
$$= 80 / (80 + 30)$$
$$= 80 / 110$$
$$= 0.7273 = 72.73\%$$

Therefore, Model A has a precision of 80% and a recall of approximately 72.73%. This means that 80% of the objects identified by Model A are correct detections, while the model successfully detects approximately 72.73% of all actual objects.

Model B

For Model B:

$$TP = 90, FP = 15, FN = 20$$

Precision:

$$\text{Precision} = TP / (TP + FP)$$
$$= 90 / (90 + 15)$$
$$= 90 / 105$$
$$= 0.8571 = 85.71\%$$

Recall:

$$\text{Recall} = TP / (TP + FN)$$
$$= 90 / (90 + 20)$$
$$= 90 / 110$$
$$= 0.8182 = 81.82\%$$

Therefore, Model B has a precision of approximately 85.71% and a recall of approximately 81.82%.

Comparison and Conclusion

When the two models are compared, Model B performs better in both precision and recall. Model A has a precision of 80% and recall of 72.73%, whereas Model B has a precision of 85.71% and recall of 81.82%. Model B also has more True Positives, with 90 compared with 80 for Model A, meaning it successfully detects more actual objects. At the same time, Model B has fewer False Positives (15 compared with 20) and fewer False Negatives (20 compared with 30). Therefore, Model B provides better overall object detection performance because it produces more correct detections, fewer incorrect detections, and misses fewer actual objects. Hence,

Model B (YOLO) is the better-performing model based on the calculated precision and recall values.

2. Motion Estimation Error Analysis

The second problem compares two motion estimation techniques based on their displacement errors over three consecutive frames. Method A produces displacement errors of 2, 3, and 2 pixels, while Method B produces errors of 3, 5, and 4 pixels. Displacement error represents the difference between the estimated movement and the actual movement. A smaller displacement error indicates that the method estimates motion more accurately. Therefore, the method with the lower average displacement error is considered more accurate for motion estimation.

Method A

The displacement errors for Method A are:

2, 3, 2 pixels

Average displacement error:

$$\text{Average Error} = (2 + 3 + 2) / 3$$

$$= 7 / 3$$

$$= 2.33 \text{ pixels}$$

Therefore, the average displacement error of Method A is 2.33 pixels.

Method B

The displacement errors for Method B are:

3, 5, 4 pixels

Average displacement error:

$$\text{Average Error} = (3 + 5 + 4) / 3$$

$$= 12 / 3$$

$$= 4 \text{ pixels}$$

Therefore, the average displacement error of Method B is 4 pixels.

Comparison

Method A has an average displacement error of 2.33 pixels, whereas Method B has an average displacement error of 4 pixels. Since a lower error indicates more accurate motion estimation, Method A performs better than Method B. Method A also has relatively consistent errors, with values of only 2, 3, and 2 pixels. In contrast, Method B has higher errors of 3, 5, and 4 pixels, with the maximum error reaching 5 pixels. This indicates that Method B deviates more from the actual motion.

Conclusion

Method A provides more accurate motion estimation because its average displacement error is only 2.33 pixels, compared with 4 pixels for Method B. The lower error means that Method A produces motion estimates closer to the actual movement. Therefore, for an application where accurate motion tracking is important, Method A is the preferred technique.

3. Depth Estimation Comparison

The third problem compares Stereo Vision and Structure-from-Motion (SfM) for estimating the depth of three objects. The actual depths are 2 m, 3 m, and 4 m. Stereo Vision estimates the depths as 2.0 m, 3.0 m, and 4.0 m, while SfM estimates them as 2.5 m, 3.8 m, and 5.0 m. The absolute error is calculated using the formula:

$$\text{Absolute Error} = |\text{Estimated Depth} - \text{Actual Depth}|$$

A smaller absolute error indicates that the depth estimation is closer to the actual value and is therefore more accurate.

Stereo Vision

Object 1

Actual depth = 2 m

Estimated depth = 2.0 m

Absolute error:

$$|2.0 - 2| = 0 \text{ m}$$

Object 2

Actual depth = 3 m

Estimated depth = 3.0 m

Absolute error:

$$|3.0 - 3| = 0 \text{ m}$$

Object 3

Actual depth = 4 m

Estimated depth = 4.0 m

Absolute error:

$$|4.0 - 4| = 0 \text{ m}$$

Therefore, the absolute errors for Stereo Vision are:

0 m, 0 m, 0 m

Average absolute error:

$$\text{Average Error} = (0 + 0 + 0) / 3$$

$$= 0 \text{ m}$$

Thus, the average absolute error of Stereo Vision is 0 m.

Structure-from-Motion (SfM)

Object 1

Actual depth = 2 m

Estimated depth = 2.5 m

Absolute error:

$$|2.5 - 2| = 0.5 \text{ m}$$

Object 2

Actual depth = 3 m

Estimated depth = 3.8 m

Absolute error:

$$|3.8 - 3| = 0.8 \text{ m}$$

Object 3

Actual depth = 4 m

Estimated depth = 5.0 m

Absolute error:

$$|5.0 - 4| = 1.0 \text{ m}$$

Therefore, the absolute errors for SfM are:

0.5 m, 0.8 m, 1.0 m

Average absolute error:

$$\text{Average Error} = (0.5 + 0.8 + 1.0) / 3$$

$$= 2.3 / 3$$

$$= 0.7667 \text{ m}$$

Therefore, the average absolute error of SfM is approximately 0.77 m.

Comparison of the Two Methods

Method	Object 1 Error	Object 2 Error	Object 3 Error	Average Absolute Error
Stereo Vision	0 m	0 m	0 m	0 m
SfM	0.5 m	0.8 m	1.0 m	0.77 m

Stereo Vision provides significantly better depth estimation in this particular dataset. Its estimated depths exactly match the actual depths for all three objects, resulting in an absolute error of 0 m for every object and an average absolute error of 0 m. In comparison, SfM produces errors of 0.5 m, 0.8 m, and 1.0 m, resulting in an average absolute error of approximately 0.77 m. Since lower absolute error indicates better accuracy, Stereo Vision is clearly more accurate for the given depth estimation problem.

Final Conclusion

Stereo Vision is the more accurate depth estimation method. It produces an average absolute error of 0 m, while SfM produces an average absolute error of approximately 0.77 m. The Stereo Vision estimates exactly match the actual depths provided in the problem, whereas the SfM estimates are consistently higher than the actual depths. Therefore, based strictly on the given data, Stereo Vision provides better depth estimation performance than Structure-from-Motion (SfM).